

AI/ML intern DAY 5

TASK 1 — ADVANCED BODY LANDMARK DETECTION OPTIMIZATION

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1. Abstract

This task focused on improving the performance and stability of Human Body Landmark Detection systems using Artificial Intelligence and Computer Vision techniques. The project involved optimizing pose estimation accuracy, reducing landmark instability, improving real-time tracking, and enhancing skeletal visualization using MediaPipe and OpenCV.

The optimized system successfully improved body tracking performance during real-time webcam processing and provided smoother landmark visualization compared to previous implementations.

2. Objective

The primary objective of this task was to optimize body landmark detection systems for better real-time AI pose estimation performance.

The task focused on:

- improving landmark detection accuracy
 - enhancing skeletal stability
 - reducing pose flickering
 - improving tracking confidence
 - optimizing AI pose processing
 - analysing body movement detection
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3. Introduction to Body Landmark Detection

Human Body Landmark Detection is an Artificial Intelligence technique used to identify and track important human body points from images and videos.

Pose estimation systems detect landmarks such as:

- shoulders
- elbows
- wrists
- hips
- knees
- ankles

The detected landmarks are connected to generate a digital skeletal representation of the human body.

These systems are widely used in:

- fitness monitoring
- sports analytics
- healthcare applications
- rehabilitation systems
- virtual fitting rooms
- gesture recognition systems

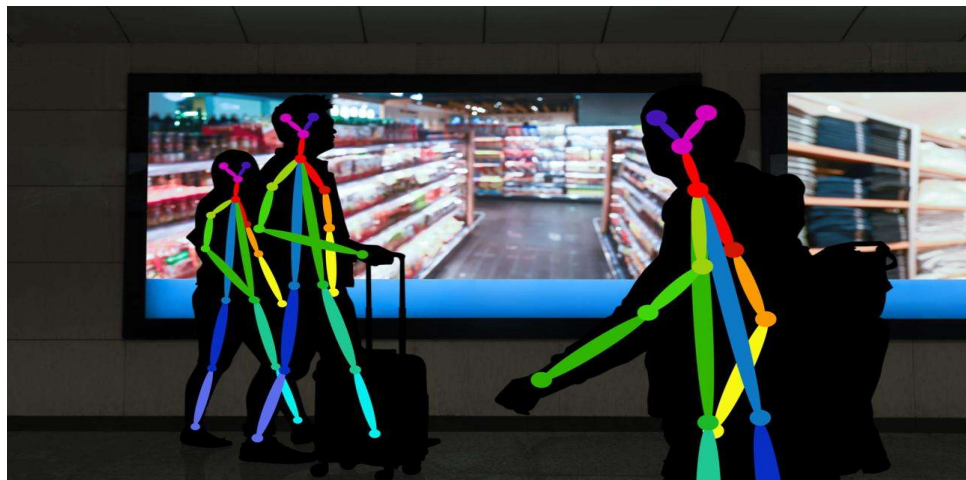


Figure 1 — AI Human Pose Landmark Detection

4. System Architecture

The optimized pose estimation system consists of multiple AI processing stages.

Module	Function
Webcam Input	Captures real-time frames
Frame Processing	Resizes and optimizes frames
RGB Conversion	Converts image format for AI processing
Pose Estimation	Detects body landmarks
Landmark Rendering	Draws skeletal visualization
Real-Time Tracking	Updates body movement dynamically

5. AI Pose Detection Workflow

Step 1 — Real-Time Webcam Capture

The webcam continuously captures video frames for AI analysis.

Step 2 — Frame Optimization

Frames are resized to improve processing performance and reduce computational load.

Step 3 — RGB Image Conversion

OpenCV converts BGR frames into RGB format because MediaPipe processes RGB images.

Step 4 — Landmark Detection

MediaPipe identifies body landmarks and predicts body posture dynamically.

Step 5 — Skeletal Visualization

Detected landmarks are connected together to create a real-time skeleton tracking system.

Step 6 — Continuous Motion Tracking

The system continuously updates body movement and pose estimation in real time.

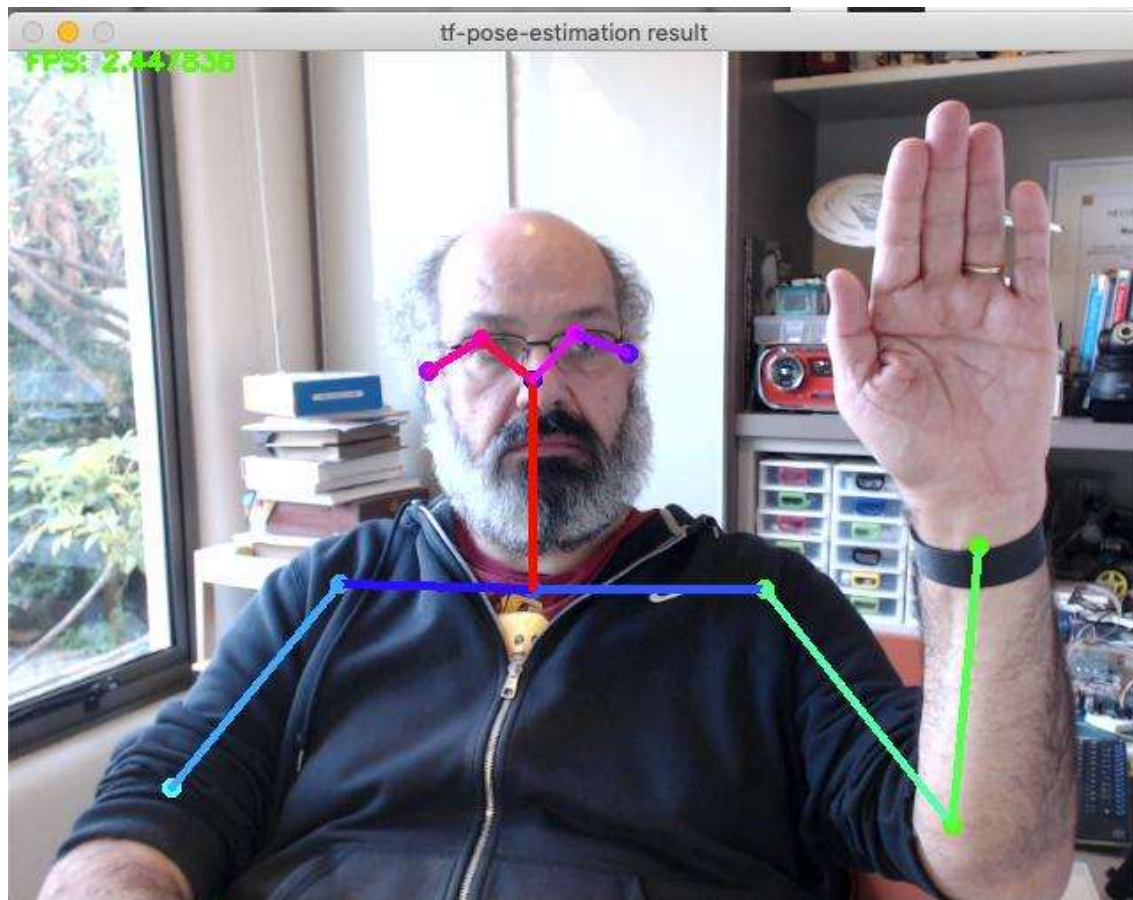


Figure 2 — Real-Time Skeletal Tracking



6. Improvements Added Compared to Previous System

Multiple enhancements were implemented compared to the earlier Day 2 pose detection model.

Previous System	Optimized System
Default pose estimation	Advanced pose optimization
Standard confidence levels	Increased detection confidence
Basic landmark rendering	Smooth skeletal tracking
Simple tracking	Improved movement stability
Minor landmark flickering	Reduced landmark instability
Default model settings	Higher model complexity

7. Optimized Python Implementation

The following Python program was developed to improve body landmark tracking performance using MediaPipe and OpenCV.

```
import cv2
import mediapipe as mp

# Initialize MediaPipe Pose
mp_pose = mp.solutions.pose
mp_draw = mp.solutions.drawing_utils

pose = mp_pose.Pose(
    static_image_mode=False,
    model_complexity=2,
    smooth_landmarks=True,
    min_detection_confidence=0.7,
    min_tracking_confidence=0.7
)

# Start Webcam
cap = cv2.VideoCapture(0)

while True:
    success, frame = cap.read()
```

```
if not success:
    break

# Resize Frame
frame = cv2.resize(frame, (960, 720))

# Convert BGR to RGB
rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

# Process Pose Detection
results = pose.process(rgb)

# Draw Pose Landmarks
if results.pose_landmarks:
    mp_draw.draw_landmarks(
        frame,
        results.pose_landmarks,
        mp_pose.POSE_CONNECTIONS,
        mp_draw.DrawingSpec(color=(0,255,0), thickness=2, circle_radius=2),
        mp_draw.DrawingSpec(color=(255,0,0), thickness=2)
    )

# Display Output
cv2.imshow("Advanced Body Landmark Detection", frame)

# Exit Key
if cv2.waitKey(1) & 0xFF == ord('q'):
    break
cap.release()
cv2.destroyAllWindows()
```

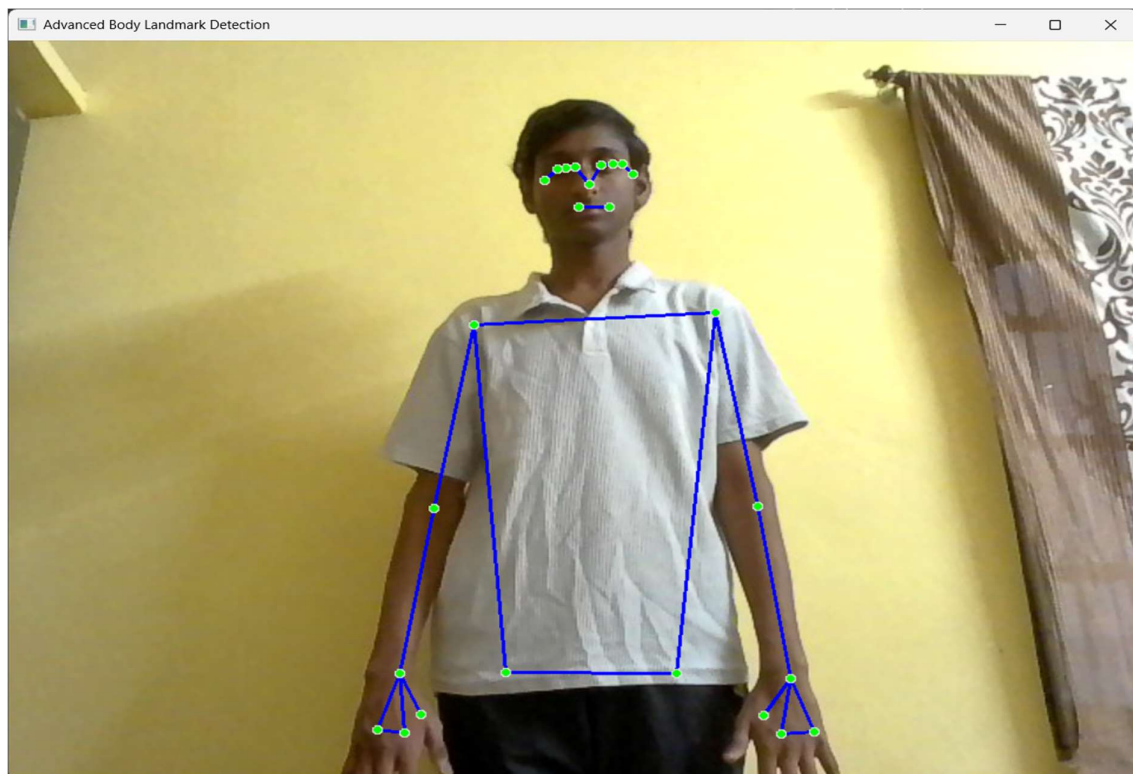
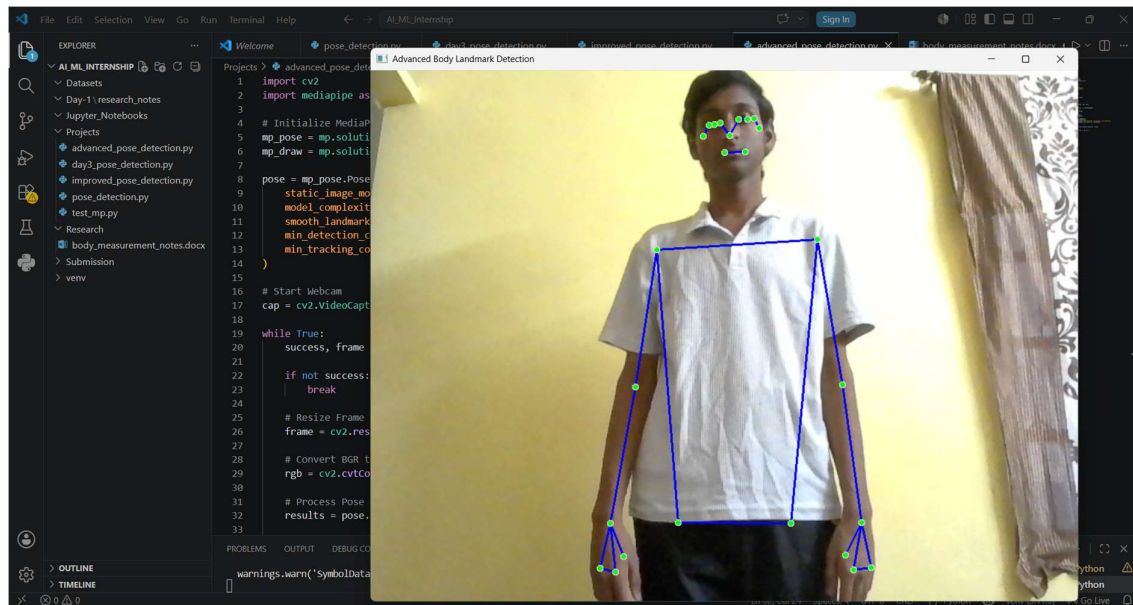



Figure 3 — Optimized Pose Detection Output

8. Technical Enhancements Implemented

Enhancement	Purpose
model_complexity = 2	Improved landmark accuracy
smooth_landmarks = True	Reduced flickering
High detection confidence	Better landmark visibility
High tracking confidence	Stable movement tracking
Frame resizing	Improved processing speed

9. System Performance Evaluation

The optimized pose estimation system was tested under multiple conditions.

Test Scenario	Performance
Full body visibility	Excellent
Hand movement tracking	Stable
Side body posture	Moderate
Fast movement	Minor tracking delay

10. Applications of Advanced Pose Detection

Advanced pose estimation systems are used in:

- sports analytics
 - healthcare monitoring
 - fitness tracking
 - rehabilitation systems
 - AI surveillance systems
 - gesture recognition
 - virtual try-on technology
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11. Challenges Faced During Optimization

Challenge	Solution Implemented
Landmark instability	Enabled smooth tracking
Low-light detection issues	Improved lighting conditions
Fast body movement	Increased tracking confidence
Frame lag	Optimized frame processing

12. Learning Outcomes

During this task, the following concepts were understood:

- advanced pose estimation
 - body landmark optimization
 - AI motion tracking
 - real-time skeletal rendering
 - computer vision workflows
 - pose estimation tuning
 - AI body analysis systems
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13. Conclusion

The optimized body landmark detection system successfully improved pose tracking accuracy and real-time skeletal visualization using Artificial Intelligence and Computer Vision technologies.

Compared to earlier implementations, the optimized system provided:

- smoother landmark tracking
- improved detection stability
- better real-time processing
- enhanced pose visualization
- reduced landmark flickering